

Python Foundations Assessment

Total Time: 60 minutes

Instructions:

- You have 60 minutes total for all 3 questions.
 - Suggested pacing: Q1 (15 min), Q2 (20 min), Q3 (25 min).
 - Don't worry about perfect code. Focus on getting a working solution.
 - If you get stuck, write down exactly what concept you're missing.
 - Do not use `eval()`, `sum()`, `statistics.mean()`, or any external libraries unless stated otherwise
 - Create a New folder in your and open it in your vs code for the exercise
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QUESTION 1 — SIMPLE CALCULATOR

 15 minutes

Problem

Create a calculator that accepts **two numbers** and an **operator**, then performs the calculation.

Support: +, -, *, /

Examples

First number: 15
Operator: *
Second number: 4
Result: 60

First number: 20
Operator: /
Second number: 0
Output: Cannot divide by zero

First number: 10
Operator: %
Second number: 2
Output: Invalid operator

Constraints

- Use `input()`
- Convert numbers to appropriate type
- Use `if / elif / else`
- Do **not** use `eval()`
- Do not import any library
- Handle division by zero

- Only +, -, *, / are valid

What This Tests

input → conversion → variables → operators → conditionals → output

QUESTION 2 — PALINDROME

 20 minutes

Problem

Write a program that determines whether a word is a palindrome.

A palindrome reads the same forwards and backwards.

Examples

Input: racecar
Output: racecar is a palindrome

Input: hello
Output: hello is not a palindrome

Constraints

- Use a loop
- Do **not** use `[::-1]`
- Do **not** use `.reverse()`
- Do not use palindrome-specific functions
- Assume lowercase letters only
- You may use `len()`
- Do not import libraries

Stretch Requirement

Don't just build the reversed string and compare. Try comparing characters from **opposite ends**:

```
r a c e c a r
↑       ↑
```

Compare pairs: `r==r`, `a==a`, `c==c`, then `e` is the middle. If any pair doesn't match, it's not a palindrome.

What This Tests

Strings, indexing, `len()`, loops, comparison, Boolean logic, translating a problem into steps.

QUESTION 3 — STUDENT GRADES

 25 minutes

This is the **real foundation stress test**.

Problem

Given a list of students and their scores, calculate each student's average and assign a grade.

Grading System

```
70–100 → A
60–69  → B
50–59  → C
45–49  → D
40–44  → E
0–39   → F
```

Starting Data

```
students = [
    ["Samuel", 80, 75, 90],
    ["David", 55, 60, 50],
    ["Mary", 35, 40, 30],
    ["John", 65, 70, 68]
]
```

Expected Output

```
Samuel - Average: 81.67 - Grade: A
David - Average: 55.00 - Grade: C
Mary - Average: 35.00 - Grade: F
John - Average: 67.67 - Grade: B
```

Constraints

- Use a loop to process every student
- Do not manually calculate each student
- Calculate the average yourself
- Do **not** use `sum()`
- Do **not** use `statistics.mean()`
- Use `if / elif / else` for grading
- Do not import libraries
- Do not hardcode final grades

- Must work if another student is added

What This Tests

Lists, loops, conditions, arithmetic, and data organization

SUBMISSION

When time is up, paste your three solutions into a reply or send them back however instructed.

Remember:

- Working solution > perfect code
- If stuck, note which concept you're missing
- You have 60 minutes total